

多自由度振動系

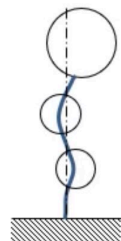
多自由度振動系



1自由度

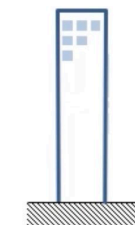


2自由度



3自由度

...

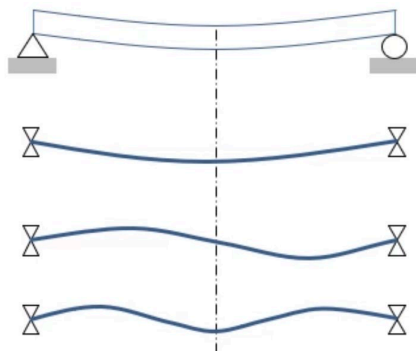


多自由度系

例) ロープ、橋、板など



連続体



連続体の振動

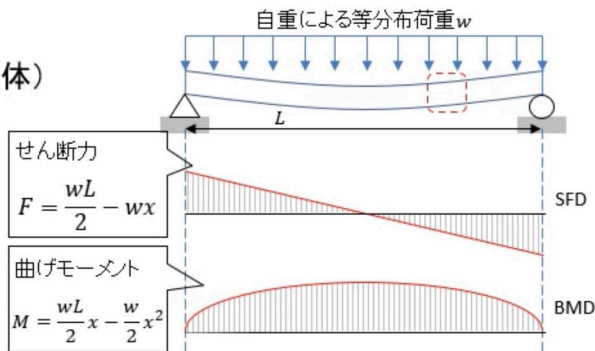
連続体の振動

板やロープなど自重のみを考えた物体(連続体)

たわみの式
$$\frac{d^2 y}{dx^2} = -\frac{M}{EI}$$

せん断力Fと曲げモーメントMとの関係は

$$F = \frac{dM}{dx} = \frac{d}{dx} \left(-EI \frac{d^2 y}{dx^2} \right) = -EI \frac{d^3 y}{dx^3}$$



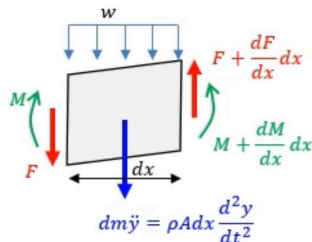
力のつり合い(下向き正)

$$F - \left(F + \frac{dF}{dx} dx \right) + \rho A dx \frac{d^2 y}{dt^2} = 0$$

$$\frac{d}{dx} \left(EI \frac{d^3 y}{dx^3} \right) dx + \rho A dx \frac{d^2 y}{dt^2} = 0$$

$$EI \frac{\partial^4 y}{\partial x^4} + \rho A \frac{\partial^2 y}{\partial t^2} = 0$$

波動方程式



この解を $y = Y(x)f(t)$

位置xの関数×時間tの関数

$$EI \frac{\partial^4 Y}{\partial x^4} f(t) + \rho A \frac{\partial^2 f}{\partial t^2} Y(x) = 0$$

変数を分離(変数分離法)

$$\frac{EI}{\rho A Y(x)} \frac{1}{\partial x^4} \frac{\partial^4 Y}{\partial x^4} = - \frac{1}{f(t)} \frac{\partial^2 f}{\partial t^2} = P^2$$

Y(x)だけの式 f(t)だけの式

Y(x)について

$$\frac{\partial^4 Y}{\partial x^4} - \frac{\rho A}{EI} P^2 Y(x) = 0$$

$$\frac{\partial^4 Y}{\partial x^4} - \xi^4 Y(x) = 0$$

$$\xi^4 = \frac{\rho A}{EI} P^2$$

Yを $Y = C e^{\lambda x}$ とおくと

$$\frac{\partial^4 Y}{\partial x^4} = \lambda^4 C e^{\lambda x}$$

$$\lambda^4 C e^{\lambda x} - \xi^4 C e^{\lambda x} = 0$$

$$\lambda^4 = \xi^4$$

$$\lambda = \pm \xi, \pm i \xi$$

$$\left\{ \begin{array}{l} -\frac{1}{f(t)} \frac{\partial^2 f}{\partial t^2} = P^2 \\ \frac{EI}{\rho A Y(x)} \frac{1}{\partial x^4} \frac{\partial^4 Y}{\partial x^4} = P^2 \end{array} \right.$$

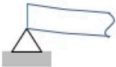
f(t)について

$$\frac{\partial^2 f}{\partial t^2} + P^2 f(t) = 0$$

2階の常微分方程式の解は

$$f(t) = B_1 \cos \omega t + B_2 \sin \omega t$$

境界条件(主なもの)

固定端	支持端	自由端	ローラ支持
			
変位 $Y = 0$ たわみ角 $\frac{dY}{dx} = 0$	変位 $Y = 0$ 曲げモーメント $\frac{d^2Y}{dx^2} = -\frac{M}{EI} = 0$	せん断力 $F = -EI \frac{d^3Y}{dx^3} = 0$ 曲げモーメント $\frac{d^2Y}{dx^2} = -\frac{M}{EI} = 0$	たわみ角 $\frac{dY}{dx} = 0$ せん断力 $F = -EI \frac{d^3Y}{dx^3} = 0$

例) 単純支持はり

$$\frac{d^2 y}{dx^2} = -\frac{M}{EI}$$



位置xに関する解

$$Y(x) = A_1 \cos \xi x + A_2 \sin \xi x + A_3 \cosh \xi x + A_4 \sinh \xi x$$

境界条件①より

$$Y(0) = A_1 + A_3 = 0$$

$$\frac{\partial^2 Y(0)}{\partial x^2} = (-A_1 + A_3)\xi^2 = 0$$

$$\left. \begin{array}{l} Y(0) = A_1 + A_3 = 0 \\ \frac{\partial^2 Y(0)}{\partial x^2} = (-A_1 + A_3)\xi^2 = 0 \end{array} \right\} \longrightarrow A_1 = A_3 = 0$$

$$Y(x) = A_2 \sin \xi x + A_4 \sinh \xi x$$

境界条件②より

$$Y(L) = A_2 \sin \xi L + A_4 \sinh \xi L = 0$$

$$\frac{\partial^2 Y(L)}{\partial x^2} = -A_2 \xi^2 \sin \xi L + A_4 \xi^2 \sinh \xi L = 0$$

$$\left. \begin{array}{l} Y(L) = A_2 \sin \xi L + A_4 \sinh \xi L = 0 \\ \frac{\partial^2 Y(L)}{\partial x^2} = -A_2 \xi^2 \sin \xi L + A_4 \xi^2 \sinh \xi L = 0 \end{array} \right\} \longrightarrow 2A_2 \sin \xi L = 0$$

$$\xi L = n\pi$$

$$2A_4 \sinh \xi L = 0$$

$$A_4 = 0$$

$$\longrightarrow Y(x) = A_2 \sin \xi x$$

$$\xi = \frac{n\pi}{L}$$

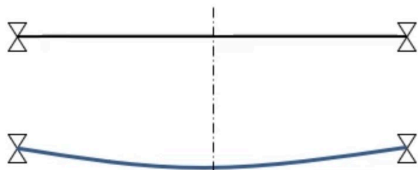
$$\xi^4 = \frac{\rho A}{EI} P^2 = \left(\frac{n\pi}{L}\right)^4$$

固有振動数

$$P = \left(\frac{n\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}$$

1次

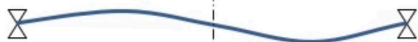
$$P = \left(\frac{\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}$$



$$Y(x) = A_2 \sin\left(\frac{\pi}{L}x\right)$$

2次

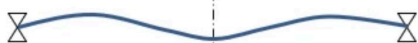
$$P = \left(\frac{2\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}$$



$$Y(x) = A_2 \sin\left(\frac{2\pi}{L}x\right)$$

3次

$$P = \left(\frac{3\pi}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}$$



$$Y(x) = A_2 \sin\left(\frac{3\pi}{L}x\right)$$